

AI Assisted Coding Assignment- 7.2

2303A51886 : 30 SAI SPURTHI. B 10.02.2026

💡 Task 1 – Runtime Error Due to Invalid Input Type

Description:

A Python program accepts user input and performs arithmetic operations. However, the program throws a

runtime error because the input is treated as a string instead of a numeric type.

Example (Buggy Code):

```
num =input("Enter a number: ")
```

```
result = num +10print(result)
```

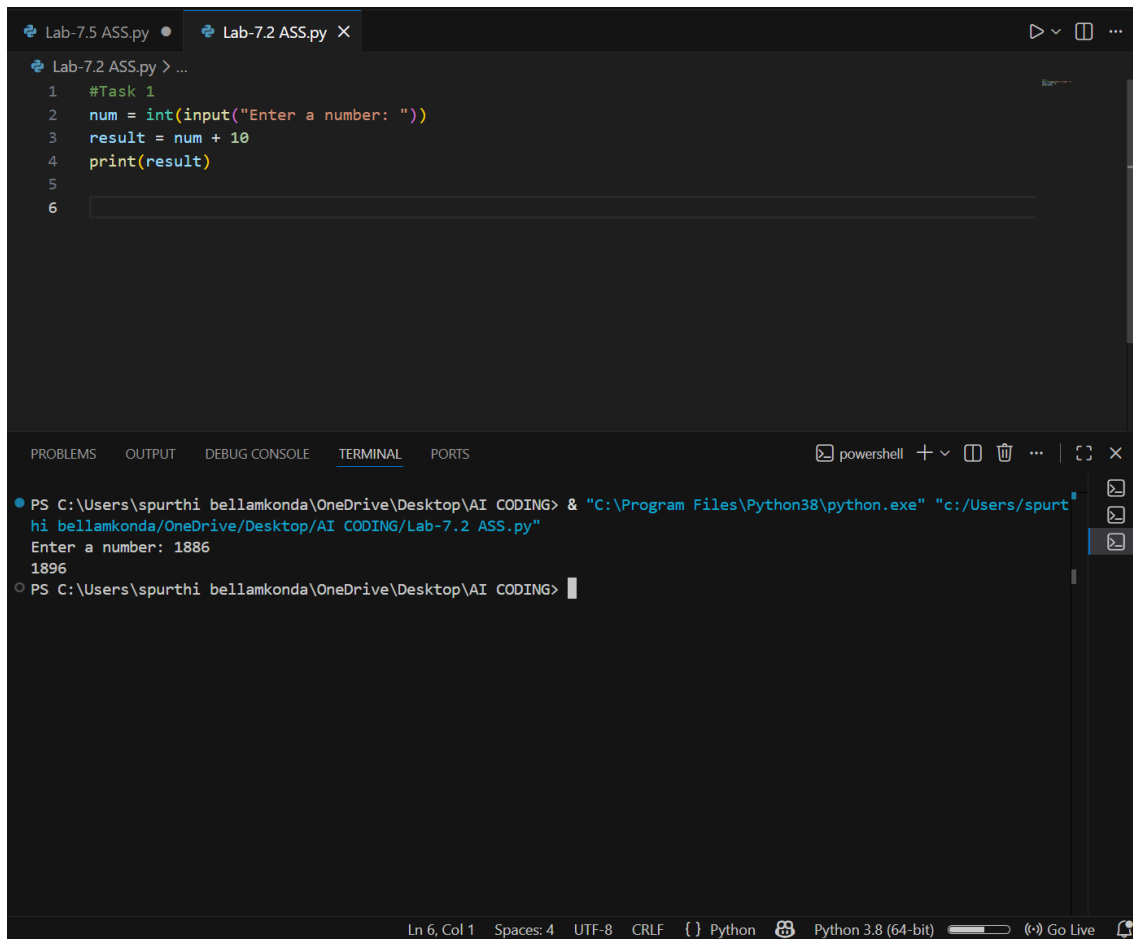
Task:

Use AI tools to identify the cause of the runtime error and modify the program so it executes correctly.

Expected Output - 1:

AI converts the input to the appropriate numeric type and eliminates the runtime error.

```
Asu72py> _
1 #Task-1
2 #Buggy Code
3 num = input("Enter a number: ")
4 result = num + 10
5 print(result)
6
7 #Fix: Identify the cause of the runtime error and modify the program and converts the input to the appropriate numeric type and eliminate the error.
8 num = int(input("Enter a number: "))
9 result = num + 10
10 print(result)
```



The screenshot shows a Visual Studio Code editor with two tabs: 'Lab-7.5 ASS.py' and 'Lab-7.2 ASS.py'. The active tab is 'Lab-7.2 ASS.py', which contains the following Python code:

```
1 #Task 1
2 num = int(input("Enter a number: "))
3 result = num + 10
4 print(result)
5
6
```

Below the editor is a terminal window with the following output:

```
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-7.2 ASS.py"
Enter a number: 1886
1896
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

The status bar at the bottom indicates the current position is 'Ln 6, Col 1', the file is encoded in 'UTF-8' with 'CRLF' line endings, and it is a 'Python' file using 'Python 3.8 (64-bit)'.

💡 Task 2 – Incorrect Function Return Value

Description:

A function is designed to calculate the square of a number, but it does not return the computed result properly.

Example (Buggy Code):

```
defsquare(n):
```

```
    result = n * n
```

Task:

Use AI assistance to analyze the function and ensure the correct value is returned.

Expected Output - 2:

AI fixes the missing return statement and the function returns the correct output.

```
Lab-7.5 ASS.py • Lab-7.2 ASS.py •
Lab-7.2 ASS.py > ...

6
7 #Task-2
8 #Buggy Code
9 def square(n):
10     result = n * n
11     #Fix:Fixes the missing return statement and the function returns the correct output.
12     def square(n):
13         result = n * n
14         return result
15     print(square(5))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-7.2 ASS.py"
Enter a number: 1886
1896
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-7.2 ASS.py"
Enter a number: 1886
1896
25
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> 
```

💡 Task 3 – IndexError in List Traversal

Description:

A Python program iterates over a list using incorrect index limits, causing an IndexError .

Example (Buggy Code):

```
numbers = [10,20,30]for i in range(0,len(numbers)+1):print(numbers[i])
```

Task:

Use AI to identify the incorrect loop boundary and correct the iteration logic.

Expected Output - 3:

AI fixes the loop condition and prevents out-of-range list access.

The screenshot shows a VS Code editor with a file named 'Lab-7.2 ASS.py'. The code in the editor is as follows:

```
18 #Task-3
19 #Buggy Code
20 #numbers = [10, 20, 30]
21 #for i in range(0, len(numbers)+1):
22 # print(numbers[i])
23 #Fix:Fixes the loop condition and prevents out-of-range list access.
24 numbers = [10, 20, 30]
25 for i in range(0, len(numbers)):
26     print(numbers[i])
```

Below the editor is a terminal window with the following output:

```
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-7.2 ASS.py"
Enter a number: 1886
1896
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-7.2 ASS.py"
Enter a number: 1886
1896
25
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-7.2 ASS.py"
Enter a number: 1886
1896
25
10
20
30
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

💡 Task 4 – Uninitialized Variable Usage

Description:

A program uses a variable in a calculation before assigning it any value.

Example (Buggy Code):

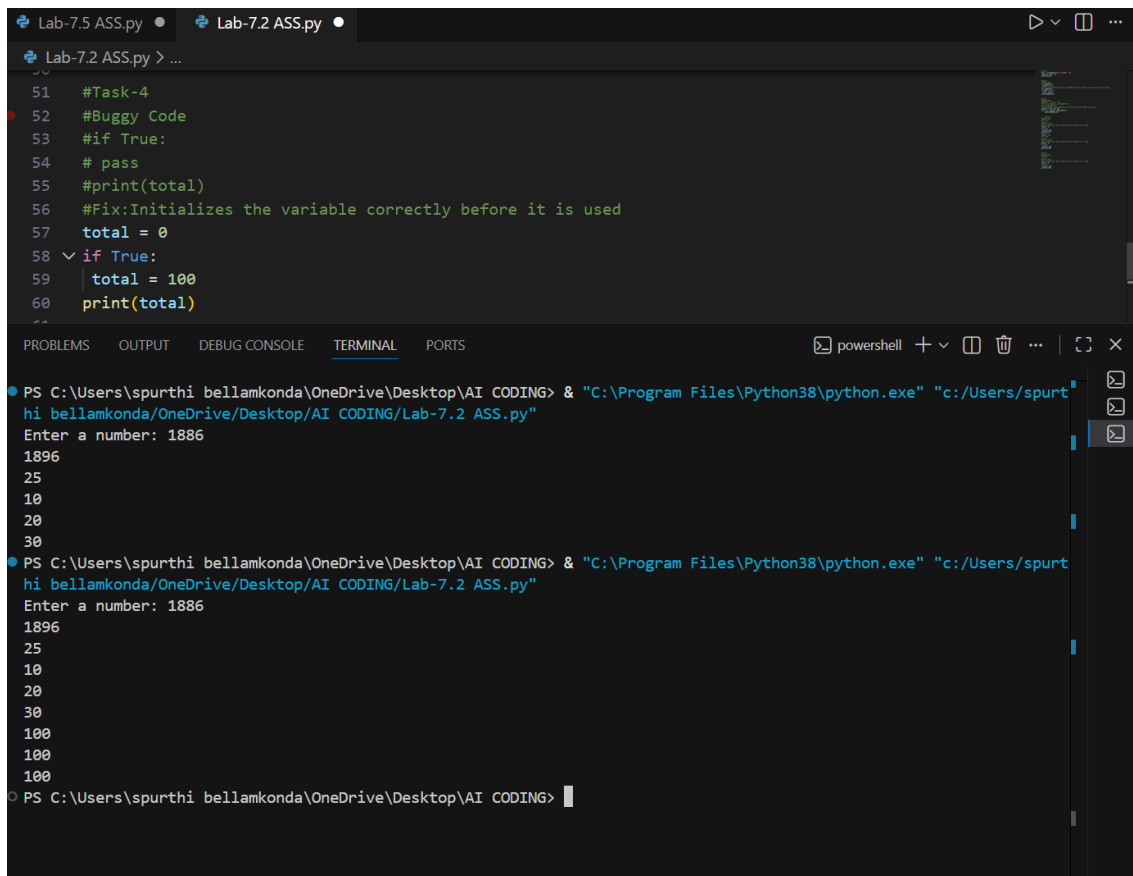
```
if True: pass
print(total)
```

Task:

Use AI tools to detect the uninitialized variable and correct the program.

Expected Output - 4:

AI initializes the variable correctly before it is used.



The screenshot shows a VS Code editor with a file named 'Lab-7.2 ASS.py'. The code is as follows:

```
51 #Task-4
52 #Buggy Code
53 #if True:
54 # pass
55 #print(total)
56 #Fix:Initializes the variable correctly before it is used
57 total = 0
58 if True:
59     total = 100
60 print(total)
```

The terminal at the bottom shows the execution of the script. It prompts 'Enter a number: 1886' and outputs the following values:

```
1896
25
10
20
30
100
100
100
```

Task 5 – Logical Error in Student Grading System

Description:

A grading program assigns incorrect grades due to improper conditional logic.

Example (Buggy Code):

```
marks =85
if marks >=90:
    grade ="A"
elif marks >=80:
    grade ="C"
else:
    grade ="B"print(grade)
```

Task:

Use AI to analyze the grading conditions and correct the logical flow.

Expected Output - 5:

AI corrects the conditional logic so grades are assigned accurately.

The image shows a Visual Studio Code editor window with two tabs: 'Lab-7.5 ASS.py' and 'Lab-7.2 ASS.py'. The active tab is 'Lab-7.2 ASS.py', which contains the following Python code:

```
63 #Task-5
64 #Buggy Code
65 #marks = 85
66 #if marks >= 90:
67 # grade = "A"
68 # elif marks >= 80:
69 #grade = "C"
70 #else:
71 #grade = "B"
72 #print(grade)
73 #Fix:Corrects the conditional logic so grades are assigned accurately.
74 marks = 85
75 if marks >= 90:
76     grade = "A"
77 elif marks >= 80:
78     grade = "B"
79 else: grade = "C"
80 print(grade)
```

Below the code editor is a terminal window with the following output:

```
30
100
100
100
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-7.2 ASS.py"
Enter a number: 1886
1896
25
10
20
30
100
100
100
B
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

The status bar at the bottom indicates the current position is Line 80, Column 13, with 4 spaces. The encoding is UTF-8, and the file type is Python. The Python version is 3.8 (64-bit).